

ActInf Textbook Group ~ Cohort 4 ~ Meeting 9 (Chapter 4 part 1)

TextbookGroup/ParrPezzuloFriston2022/Cohort_4 | Cohort 4 ~ Meeting 9 (Chapter 4 part 1) | 56:39

Hi, everyone.

It's August 1st, and we're in Cohort 4, having our first discussion on Chapter 4.

So before we go into more detail on Chapter 4 and everything, does anyone have any just overall thoughts or comments on Chapter 4?

What did they expect it would be delivering as a chapter?

How was it?

Or what was any aspect that people remembered or anything?

I found this chapter to be rather mathematically dense, I suppose, and the authors do warn of that at the beginning of the chapter.

I mean, I think it's great so far as it starts to introduce how they're using Bayes' theorem, how they kind of exploit Jensen's inequality to move in the direction of calculating free energy, using logarithmic functions in order to do that.

You know, the initial models of the static perceptual tasks also seems like a really nice introduction to moving on to POMDPs, as well as the continuous models used in active inference.

So, you know, in a way it felt like it kind of started with nice first principles and built up from there.

But that said, I can start to see maybe the necessity of having a math learning group for some folks here, given this chapter.

Thanks.

Yeah.

Anyone else?

Any, like, just overall thoughts or?

Well, I would just totally agree 100% with what Andrew shared, basically.

It's just similar thoughts.

Yeah.

Yeah.

I mean, basically the, the, the, the maths, uh, are very well explained and it goes gradually, but I think it would be beneficial to,

yeah, to go even deeper and slower through them, maybe separately.

Definitely.

Any other thoughts people have?

Yeah.

Yeah.

Darius.

So then anyone else?

Yeah.

Daniel, it's less a point about the chapter, although I agree with the previous sentiments that it is mathematically dense.

It's more of a point about the math learning group.

I think myself and several others just having trouble locating it, um, in terms of where it actually is on the discord.

So the past couple of weeks I've been trying to get into some of the voice chaps and it doesn't, there doesn't seem to be anything kind of happening on them.

So there might be a bit of confusion.

Yeah.

Thanks.

It is, uh, people can use the general voice chat, just the top voice channel.

And, uh, you know, if there's no one there, just, just still jump in and stick in there.

Okay.

Yeah.

Yeah.

Yeah.

Yeah.

Yeah.

Yeah.

Yeah.

Yeah.

Yeah.

Yeah.

Yeah.

Yeah.

Yeah.

Yeah.

Yeah.

Yeah.

Yeah.

Yeah.

Yeah.

Yeah.

Yeah.

Yeah.

Yeah.

Yeah.

Yeah.

Yeah.

I'll try to join when I can, but it can, I mean, there's so, so much work on the more introductory side, like kind of just going from the fundamentals on through all the work that Jonathan and Ali and others are doing too.

So there's no lack of things to do in math learning group.

I hope people do like connect on it.

Younghoon?

Wait, I don't hear you.

I can see you're unmuted.

Maybe frozen though.

Can anyone else hear or no?

Okay.

Okay, Younghoon, when you're back, feel free to continue.

Any other general thoughts on four or anything writing in the chat?

Okay.

Just another small specific observation that's a, as a kind of a practicing entry-level data scientist, the way, something that helped me understand the static perceptual models earlier on is that it most feels like it's just describing something like the process of binary classification. One of the models in the figure 4.2, the very first one.

You have the example of like your prior being prevalence of a disease.

Is there a disease?

Yes or no.

You have a true positive rate, true negative rate.

You end up with the result of your test.

I found that to be rather interesting and useful.

Just are we trying to predict does someone have a disease or not?

There's more that goes into that step by step.

Yeah.

Thanks.

Thanks.

That's one of the classic Bayesian examples.

Like there's a test that's 99% accurate and there's a disease that one in 10,000 people have.

What is the probability that you have the disease if you got a positive test result?

And the kind of snap answer type one answer is 99% because if it was positive and it's 99% accurate, then how could it not be 99%?

But when you actually unpack the Bayes equation, it's actually not 99%.

It's actually still unlikely with those parameters.

Young Hoon?

Oh yeah.

Sorry about it.

Go for it.

So I find these mathematical equations in chapter four to be a little bit ambiguous, especially like notations or like dimensions or whether it's a vector or whether it's a scalar or etc.

So depending on the context, like I find especially difficulty to understand the exact equation when there are some functions called like categorical function.

So I just accepted the concept, but just wanted to make it more clarified.

So I actually uploaded some PDF and questions to the chapter four because I got stuck with some

derivations of it.

So that might be in the court for chapter four questions and you can look it up later on.

Awesome.

Awesome.

We'll definitely look at it.

And also here I see one this Overleaf link.

This was prepared.

Well, I guess he just added this in.

This is from Jonathan shock.

Yes, I found it to be very helpful.

That's that's awesome.

Thank you, Jonathan.

Yeah, I mean, there's even already there's all there's so many pieces of material.

It's it's you know, it's like a second level question how we how we organize things so that they're discoverable and kind of annotated so that somebody who's looking for like this level.

Of of detail can can connect with this type of material.

Let's just read his basic remarks.

Free energy and expected free energy are introduced almost in parallel, but it's not clear which one should be used when or exactly why one needs both of them.

This is relatively simple that the free energy is used to improve the model itself.

Whereas the expected free energy is used to make decisions about which is the best action to take, which will lead to states which are preferable and states which lead to more epistemic value, which will themselves allow one to update the free energy.

Awesome.

Yeah, this looks really good.

Yeah, I'm.

Any other like just thought or question or area in for that people want to jump to.

Young again.

Oh, yeah, so I I'm not sure whether it's a good explanation or I'm not sure whether it's a correct intuition.

Young again, I think, for me, it feels like like comparing with reinforcement learning.

Reducing the free energy is something a little bit related to offline learning.

While.

Minimizing the expected free energy seems like to be related to online learning, so you use choose decisions, and then you try to reduce the free energy online.

Or the expected free energy while you reduce the variation of free energy offline.

So I'm not I'm not saying that those are equivalent, but it seems to me that they are somehow related.

Maybe just summarize what is online offline learning.

So I'm not really a master of reinforcement learning, but to it.

Based on what I understood about it is that online learning is learning while you're doing the action.

So offline learning is something more like you learn the value of something based on the state trajectories that you already visited.

So it's more like learning week by replay.

I would say it can if anyone's thinks that it's I'm explaining in a very ambiguous term, or if I'm wrong, just let me know.

Oh yeah, definitely an area to look into Darius.

I think what might be useful for me and perhaps others is just getting a bit more clarity on exactly how we define the generative model in active inference.

So on one side, and you hit and you see it expressed in this chapter that comprises the prior and the likelihood term in Bayes' equation.

But then on the other, I've seen other theorists talk about how it's actually basically the entire conditionally dependent dynamics across the Markov blanket.

So includes the external dynamics. It's not just what's internally encoded in the internal states of the particle.

So unless that's just a continuing conflict or continuing debates within the active inference community, it would be good just to get a little bit of clarity because I think that also helps us explain exactly what is minimizing free energy and whether we can talk about agent having a generative model or being a generative model.

Yes.

It's a great point.

And it has come up before.

So in fact, there are about one and a half definitions of what the generative model is, and they're not identical. They're kind of nested concepts.

And it certainly does matter.

And so I can understand why it's a point of confusion.

So in the 2022 textbook that we're reading right now, generative model is very, very clearly differentiated from the generative process.

So like when we're looking at the figures, it's like, here's the generative model.

That's like the agent.

Sometimes we would call that the particular states because it constitutes the whole particle.

Like just like if we were talking about a dust particle Brownian motion.

Now we're talking about the cognitive particle.

So that includes the blanket states and the internal states.

And so that is sometimes called a generative model.

And the generative model as it's framed in the textbook is to understand or to make this model of the generative process, which is the actual process that gives rise to observations.

So this distinction between generative process and generative model is is heavily used in the textbook.

However, more recently, some authors have have taken a slightly different approach or definition.

So as you mentioned, and for example, in in this representation from Maxwell on, I guess, a month ago or something around then, here generative model is used to describe the entire particular partition, all the

states, including the external states.

And in, in, in, in, in here's all these question.

Earlier, there was an emphasis on distinguishing between generative process and generative model, but in this figure figure one, which you show in the slides,

we don't see any generative process.

So where does generative process fit in the diagram?

What is the distinction between the concepts?

And yeah, I'll, I'll, I'll, I'll put these into the code as well.

At least Maxwell is not using the generative model generative process distinction as much.

So that's how I kind of reconcile this.

Whatever you want to call different parts of the particular partition, you need all four of them to be defined in the model.

So whether you just call them, you know, whether you buy them a la carte, or whether you buy them in a bundle, all the particular states, all the states need to be described.

Otherwise, you're going to have a null variable.

And now there's several, we could, you know, we could talk about, you know, you have your friends who like this music, and you have your friends who like this food.

And so there, there's, there's different nested levels of describing the friend group with these four friends.

And so there's, there's different states.

One of them is the way that the textbook describes with the generative process, external only, and the generative model with these three on the right, blanket and internal.

Another group that we talk about sometimes is just the blanket states.

And of course, states are not intrinsically blanket states, because every state is blanket is something else and so on, but in this setting blanket.

And then another set of states that we talk about sometimes are the autonomous states, which is just the internal states and the active states.

Autonomous states are the two things that can be changed to reduce free energy.

Change your mind, learning, perception, attention, etc. And action, change the world.

So that's the no thumb on the scale.

So whether or not you focus on these three on the right and call it a generative model, and this is a generative process, or whether you call all four the generative model.

At the end of the day, the variables are defined the same way.

And the variational and the expected free energy are still regarding the autonomous states with respect to the sensory states.

And so generative model and generative process is a little bit of like a linguistic descriptor or handle.

But it doesn't change anything to give a different English name to describe a subset of nodes.

So I don't think it's an issue. However, it is confusing.

It is confusing.

I just get one more point of clarification on that. Yeah, yeah.

You go back to that diagram that would be quite useful, actually.

Here?

So yeah, that's perfect. In terms of self-evidencing, then, if we take Maxwell's idea that it's the generative model is the system, the across the Markov blanket.

Do we talk, could one consider that system to be self-evidencing?

Or is it only the particle which excludes the external states, which is self-evidencing?

That's a great question. I think also calls back to your rock climbing example.

Yeah, I agree.

Okay, the rock climber is doing something cognitive about the wall, but is the wall doing something cognitive about the climber? Or is there some...

Are we considering the entire system as doing some kind of gradient descent?

Yeah, that's...

Yeah, that's... it's a good question.

Either side of the blanket, I believe, may be seen as doing some kind of self-evidencing.

Now, in the case of one active, adaptive, sentient entity and one inert one, the inert entity's self-evidencing is going to be trivial.

So just because there's self-evidencing on both sides of the blanket doesn't mean it's the same kind of self-evidencing.

But you can also make situations where it is the same kind of self-evidencing across both sides, like in the bird song example.

Does that... do two... when two sentient agents are engaged in self-evidencing, does that constitute higher level of self-evidencing at the entire system level description?

Possibly.

I mean, if they're both reducing free energy, now what if one is reducing free energy and that is creating an increase in free energy for the other?

Could that increase the net free energy on the system? I don't know.

Or is it always the case that it... that the ball has to roll downhill and the more massive, cognitively massive self-evidencer takes a stronger hand and that makes it move down on average?

Also, I don't know.

Andrew?

Andrew?

I don't want to pull too far away from the kind of current formulation that's being discussed here, but something that has been useful for me, and it... I think... please correct me if I'm misstating this, but with differentiating the generative model and the generative process, in part, is that the generative model is presumed to have

like, you know,

something like preferences or the...

something like preferences or the...

its own C matrix or its own homeostatic set points, which I believe the generative process does not have.

So we have like from chapter 3 and so on, the generative process or the environment of the agent, the generative model, is presumed to have like, you know, dissipative forces, entropy, and so on.

And so the generative model is kind of acting against those in order to realize its own expectations or preferences.

That is, I want to survive as a generative model.

The environment around me has no set of direct preferences saying like, Andrew needs to survive. I'm the one who's saying that. And so despite the fact that I'm attempting to infer or approximate the external states that I can get around in the world, like I'm also kind of acting against it, or at least, you know, in a way that doesn't...

that's never gonna exactly equate to the generative process around me.

And this is where the map does not equal the territory kind of formulation starts to come into play and becomes all the more important.

Is any of that tracked?

Yeah, that's a great point.

Certainly map doesn't have to be the territory even structurally like the generative model.

Too cold, just right, too hot, three state categorical. That's the categorical distribution.

Or it could be a continuous variable 0 to 1 or 0 to 100.

And then outside it could be a totally different structure.

And so the generative process, I believe it's... or if you just call it, you know, the process generating observations, that could just be a sine wave, or it could be a square wave, or it could be a deterministic system.

And so that makes it very clear, like you said, that when we're talking about the generative model of the agents, we're talking about A, B, C, D, E.

Those are the variables that have to be defined.

That's figure 4.3.

Basically, whether we're working in continuous time or in a discrete time, these are the variables that have to be defined or accounted for in the generative model.

This doesn't describe how those observations are generated.

So this is kind of like just the agent lifted from the environment.

And so it still is help, and it's the one that's exerting preference and engaging in sentient behavior and all of that.

And then it's like helpful, again, whether you call it generative process or underlying process generating the observations or the components of the system that generate the agent's observations.

Just to make it clear that we're not always in a situation where it's two active entities talking to each other, like the birdsong example, but rather we can be in a situation where, like the rock climber that Darius brought up, where you have like this figure 4.3 agent.

And then there can be an articulation with a generative process again or whatever you want to call it that is not doing like policy selection.

Except in a trivial way.

The instrumentalist flavor is we can model what the brain is doing with Bayesian statistics, which is to say like updating.

What kinds of Bayesian statistics?

Well, the inbound sense making signal processing side, and then the outbound action selection control theory side.

Two operations of kind of the general cybernetic entity, and we can use Bayesian models, Bayesian methods there.

That's the problem of inferring states of the world perception inbound sense making and inferring the course of action action selection outbound planning.

Exact inference is generally computationally intractable or infeasible.

And separate from that limitation in psychology often what's being considered as bounded rationality or like constraint limited cognition.

So how can we respect that we're modeling bounded rationality, approximate rationality, limited finite systems, and kind of at the same time move beyond this limitation of exact Bayes being intractable?

Like in the frog jumping out of the hand example.

It was just some small matrices to multiply, and so the exact Bayesian calculation was not out of our grasp.

However, if we were going to be doing some kind of large scale data analysis, then, of course, we would need to do some approximation methods.

So there's two large, just speaking broadly, there's two ways to approach this kind of approximate Bayesian calculation.

And they're related, and there's many, many sub flavors, and Livestream 52 is a good place to learn more about this.

But there's sampling methods and variational methods.

Sampling based methods or Monte Carlo methods that came from the card games that were played in Monte Carlo.

And it was like, well, like what's the probability of getting two twos, four red cards, and one clubs?

I mean, you could write an expression and try to analytically solve that.

Or the sampling based approach, which has become called the Monte Carlo approach, is where you just draw hands of cards, and then you draw a million hands, and you say, okay, then there was three.

So then in the end, you say, my best estimate is this happens three out of a million times.

Now, you can probably see a lot of limitations, but advantages of sampling.

On one hand, it allows you to approximate arbitrary, potentially analytically unruly distributions.

So even where you don't have a nice, smooth function, or you don't know a lot about the system, sampling may work quite well.

On the other hand, you never really know if you're sampling appropriately or if you've sampled to completion.

The other methods that are utilized, and also this cannot be done incrementally very well at all.

You need to curate many, many, many samples in order to make an estimate.

And where state spaces are large, like the state spaces of all genomic diversity for an animal with a billion base pairs in its genome, there's going to be certain things where it's just not really tractable.

In contrast, variational methods are an information theory-driven approach that is going to use all the tools

that we'll come to know and love, like the KL divergence and the variational free energy.

So that's kind of like why we're taking this path at all, because exact bays would be epic, but it's intractable.

Exact bays would allow us to compute the minimum surprise model, which would be the maximum evidence model.

We're not going to be able to compute exact bays in general.

How can we approximate it?

Well, with sampling and or variational.

Sampling has some situational advantages, but variational is a much, much, much more flexible approach.

It's a lot more tractable.

It's incrementally optimizable.

We can reduce the KL divergence and improve our model fit incrementally and kind of indubitably.

Like we will always make the fit better.

If we reduce the KL divergence, we're making the fit better.

Now, that doesn't mean that the final fit or anywhere along the way is adequate for your business or your company or whatever.

You could be moving a Gaussian distribution to its best possible fit location in the middle of a bimodal distribution and still be totally missing the point.

So this isn't to say like it works.

It's just to say the ball rolls downhill.

So that's like the big picture on why variational methods are being used.

So chapter four is going to describe generative models, which are using Bayesian networks, using Bayesian methods to describe cognitive systems, sense making an action.

And then the variational methods are going to support us doing optimization and inference with these models, approaching exact Bayes asymptotically.

So, but instead of focusing on surprise minimization itself, which we're not going to be able to directly calculate, we're going to be able to calculate something that is bounding surprise.

And that's going to be the variational free energy.

So, we're going to be able to calculate the variance.

And in figure 2.4, we kind of saw what that looked like as well.

Variational free energy, equation 2.5, the distributions and the data.

And if we can shrink that divergence, we're getting ratcheting closer to surprise.

So that's why it's possible to use variational free energy to incrementally optimize and update and learn and use that.

Tractably.

Jensen's inequality would be awesome to have more of a, like an unpacking on this because this is kind of, it's very provocative in general.

But suffice to say that because of the curvature of the natural log function or any other function that's concave down like that,

we're then able to use that property of this curvature type function and properties of logs in order to generate

an inequality.

Such that the log of the expectation is always greater than the expectation of the log.

So it's kind of like an application of Jensen's inequality for concave down functions in this setting to heuristic bounding of what we would really want to know.

So there's some further rearranging.

Again, I'll review later.

Perhaps Jonathan's writing makes this clear or other people can fill in some dots.

But where this is all kind of like more detail on equation 2.5, just helping us understand variational free energy as kind of like the Bayesian optimal way to fit.

Where's Jonathan's writings on the chapter four page?

It is here and I'll put it in the chat.

Thank you, Kate.

So that was 4.2.

Section 4.2 was just saying Bayesian inference, frog jumping out of the hands.

So if you can calculate it exactly, go for it.

No one's going to stop you.

But in situations where we want to do an approximatable heuristic and one that's incrementally optimizable, we can use free energy.

This is also known as energy based methods or energy based learning or variational methods going back to decades ago.

So this is not invented by active inference at all.

It's also called the evidence lower bound in Bayesian statistics.

So this is just applying that methodology approximate Bayesian computation with a variational flavor, not the sampling flavor to, in this case, the question of perception.

Now, you know, having made the software update from just pure exact Bayes to approximate Bayes.

Now we can calculate the free energy for a generative model.

So whereas an exact Bayes, we would be talking about getting the maximum evidence model, minimum surprise model, plug and chug.

Now we're going to get the minimum free energy model.

Which is going to be converging towards that minimum surprise maximum evidence.

The book is going to return and continue to juxtapose in contrast to major architectural decisions that are made for dynamical models.

Now, not all models are dynamical.

Not all models involve something changing through time.

But in general, when you are modeling something that's changing through time, you can either have a continuous time model or a discrete time model.

Discretization sometimes still has to come into play for continuous models and so on.

But this is like the two broad approaches to dealing with dynamical systems, continuous time, discrete time.

And that's why figure 4.3, we're always coming back to again and again with a discrete time and with a

continuous time.

Because it's like a Rosetta Stone that's like saying whether you take this approach to modeling time or that approach, there are some consequences.

And there's different settings where they may be more or less appropriate.

But we're still talking about the same infrastructure and logic for generative models.

But before building up to 4.3, which has this like downstairs perception, passive perceptual inference component, and then an upstairs decision making component.

And analogously for continuous time.

Figure 4.2 is just getting us familiar with what these causal graph structures are and do.

And so it's a fun exercise or question like to first off draw different situations that you're interested in or thinking of as like Bayesian causal graphs.

And then also to look at the ones provided here and then think about what real world situations map onto these.

Like maybe here's was there a lightning strike or not, unobserved.

But then we observe the light and the sound.

Or there could be other examples, but this is just kind of like bring the level of rigor to causal flow chart mapping.

And these are really extensible and composable.

Like you can connect them in many different ways.

And given their connectivity, you've defined like the necessary and sufficient information for that model.

It's not like, oh, and there's a secret connection from, you know, this one back to that one.

It's like what's written is there and included and calculated with standard methods.

What's not included isn't included.

It'd be like a calculator is only going to add what you tell it to add.

So inference is going to proceed along this infrastructure without some other secret third thing influencing it.

So they talk a bit more about just what these how to read the relationships between different variables.

Then they get to figure 4.3, which is the Rosetta Stone figure between the discrete time and the continuous time.

Anyone want to give a thought or a question on figure 4.3?

Well, we're going to return to the contrasts of discrete time and continuous time multiple times.

It's going to come back most fully in chapter seven and chapter eight, which are wholly dedicated to discrete and continuous time.

But in this section, it's going to foreshadow and discuss those two different types. Darius?

Presumably, this model is a kind of simplification of what might be going on in the cognitive creature.

I because I presume there are some backward loops in which the observations that you make are informing your action policy, hence the action perception loop.

Similarly, I was also I also make the presumption that the state dependencies have to be informed by prior action policies and beliefs.

So priors about these. So I presume it's just this architecture needs to be built out a little bit more or else it's

not really governed by any particular tell us by any particular incentive.

Yeah, great points.

There is a prior that helps get the ball rolling.

And so that prior kind of starts the chain.

But then in subsequent time steps, the action and the unfolding of the situation at the prior time point sets how things are.

And certainly these are like these are like.

Like a frame, it does not have more advanced cognitive features like attention, anticipation.

So on those things get get layered in.

But this is kind of like the kernel or like the motif that we then can can elaborate and bring richness to.

And so that's like why sometimes there's an adjective in front of active inference like sophisticated active inference or deep active inference affective active inference.

And those adjectives call attention to certain structural motifs that are being brought into play.

And this is kind of like this is like the lowest common denominator.

Because we have a prior on hidden state.

What's a prior on the temperature of the room?

And then we have the temperature of the room changing through time and the thermometer readings emitted at each time.

That's the downstairs passive inference bit.

D is a prior hidden state real temperature.

B describes how temperatures change through time.

A describes the relationship between temperature emitting observations or observations inferring back to temperature.

Downstairs looking like an E facing down passive inference.

Where do we embed action in the partially observable Markov decision process?

Well, the partially observable part is on the bottom.

Because we have parts that are observable and parts that aren't.

So it's partially observable.

The Markovian property through time is that the past only influences the future through the present.

Partially observable Markov.

And then the last piece is the decision process.

π is here policy.

It's intervening in how the world changes through time.

And what guides policy selection?

Well, you could make a POMDP that has a stochastic policy selection or has reward driven policy selection.

Or as we're going to see, we utilize expected free energy to guide policy selection.

So this is kind of bringing the well-known POMDP format to bear on our setting.

Younghoon?

So I'm a bit confused about this generative model because as you said, it's a generative model.

So do you, does this model actually mean that it's a con?

We're doing some kind of calling process about this because to what I understood is that number one, number two, and number three, which is like the maybe the number three, which should be located as the transition metrics.

So that's a policy.

So that's something that I'm being confused.

So that's something that I'm being confused.

So because if you're saying generative model, it means it sounds to me like something that is happening in our brain.

But what I understood was that this is a diagram about the external world itself.

Great, great question.

So in fact, we see two representations.

One of them is the particular partition looking representation with the four internal and external blanket states. Here, we don't see external states at all.

And it's not even shown like where they connect in.

And this is actually all happening within the cognitive model of the agent.

I think the figure that clarifies this the most is 9.1.

So now we're pulling back to the behavioral experimenters lab.

So here's going to be the rat in the maze.

We are using the POMDP architecture to make a model of the rats beliefs about temperature changing through time and its beliefs about how policy influences that its beliefs about how temperature and thermometer are linked.

So the POMDP is describing the agent's parameterization.

It is not the actual true unfolding of the system.

Here, the observations are being actually the generative process is in this outer layer.

If we want to think about it that way.

So yes, the B matrix is something that the agent learns about how states change through time and how its actions intervene.

But that's not actually how things change through time.

That's tracked by another variable.

Yeah.

So does it mean that those the P function annotation P should be actually Q because it's an approximation, isn't it?

So like I don't really remember which chapter it was, but maybe chapter 2 the diagram actually kind of shows that P is actually an annotation for the notation for the probability happening in the real external world while Q is something that we approximate.

So basically I think it's better.

Yes, I would like to see someone else also check on that.

However, yeah, I believe it makes sense to think about these as the belief distributions of the agent and hence utilize Q , whereas here P is just being used as like a generic statistical distribution.

But I am not 100% confident on that, but I agree.

And that is how we think about it from the agent's perspective.

And I agree that the Q 's and the P 's, you know, mind your Q 's and P 's, it does get confusing because it's like, well, wait, but if we know anything about P , why are we not just like using that information?

Like, why are we even messing around with Q if we can contrast it with P ?

So these like minor notation ambiguity is actually making me kind of confused about the equations.

So just wanted to clarify.

Well, in the.

I would love to help connect the ontology terms, which then we can support with translations and everything to the equations and to the different representations, because I agree.

And the equations work out and then second layer renderings or schematics often include or exclude things that that reduce the clarity of what they're representing.

Well, in the rest of four discrete time is explored more.

That's basically more math about the top part of figure four, three, section four, five continuous time, more math on the bottom of figure four, three.

Some extremely spicy and interesting topics with Markov blankets and message passing.

Some.

Some representations of message passing.

Generalized coordinates of motion.

And some visualizations about that connecting it to Taylor series expansion.

Laplace approximation, another tool in the toolbox to help approximate.

A given inference problem by using a quadratic expansion around the mode.

More message passing.

And that's the end of four.

So definitely it's a dense chapter.

Most of the density is in this second half of four, where it's just giving a lot more detail on discrete time and on continuous time.

And the big picture is we're introducing the kernel or the minimal motif and some of the mechanics associated with using discrete time and continuous time models.

So next time we will turn to the cohort for questions.

So it'd be awesome if people add questions and and vote them and we'll come back and take it from there.

Okay.

Any last thoughts?

Okay.

Thank you.

Farewell, everybody.

Bye.

Thank you.

Thank you.

Thank you.

Thank you.

Farewell, everybody.

Bye.

Thank you.